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Corso di Laurea in Ingegneria dell'Automazione e Robotica

Progetto di Identificazione e Controllo Ottimo

Identification and optimal control of an electrical storage system for microgrids with renewables

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Abstract

The key idea of this work is to develop an MPC controller to determine the optimal scheduling for the storage power of an Energy Management System characterized by a two-level architecture: the higher level, based on a receding horizon control scheme, that optimally schedules the operation of the storage by using information on radiation from a forecast service provider while the lower level implements an heuristic procedure (if-then) on a low-cost local controller, in order to perform corrective actions. The proposed architecture is based on a real case study [1].

1 Introduction

The urgent need to cut down on pollutant emissions and energy consumption has been leading to the use of distributed renewable energy systems. In this changing scenario, EMSs (Energy Management Systems) have become essential for effectively integrating intermittent renewable sources and electrical grid. Specifically, the installation of microgrid with storage system and photovoltaic (PV) have become a pivotal setup for residential and commercial users [2]. However, managing storage systems paired with PV installations has led to the need to have low-cost EMS. The high cost of the storage provides a challenge for an optimal scheduling that allows to maximize the profit, in particular, these new low-cost EMSs should be able to satisfy electrical demands while finding an optimal schedule of the storage to minimize the costs. To this end, the approaches presented in the literature can be divided into two different categories: heuristic-based and optimization-based approaches. The first one's, based on the instantaneous values of the power produced by the different units, exploit simple rules, which are frequently implemented in commercial products like simple control units and PLCs. The second one necessitates a higher computational burden, aimed at minimizing a cost function (having an economic and/or an environmental objective), or at tracking a predetermined power profile.

2 Storage Model Identification

In this section a Na-Ni battery model has been made with the purpose of including it within the optimization architecture. Storage systems are complex devices characterized by batteries with different chemical characteristics power electronic equipment and local control system as shown in Fig.1. This is why it's necessary to derive a precise model for the considered study.

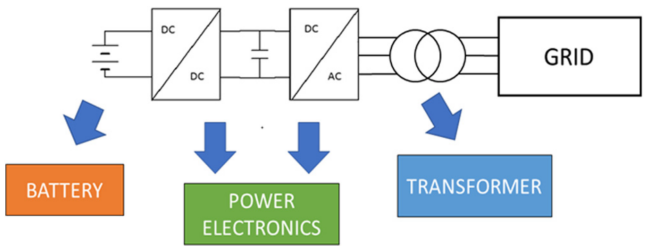


Figure 1: Schematic representation of a storage system

These are high temperatures batteries operating at around 300°C without needing cooling, since they work in an adiabatic case and thus, they are unaffected by environmental conditions. Moreover, they don't use rare or toxic materials making the disposal of the batteries easier, in addition to the fact that they require low maintenance. The following model is determined by using real data obtained through tests reported both for charging and discharging phases in [1]. The tests have been conducted by imposing power set points equal to -15 kW and -30 kW for the charging phase while the discharge phase are set to 15 kW and 30 kW. It must be pointed out that the actual power absorbed by the storage system during the charging phase or delivered by it during the discharging also depends on the current state of charge of the battery (SOC).

2.1 The considered classes of storage models

Model structure (charging)
By the patterns registered during the experimental tests, a first structure of the model to represent the dependence of the actual absorbed power from the state of charge has been selected (1)

$$P_s(SOC) = \begin{cases} P_s^* * 0.98 \quad if \quad SOC \ \leq a_1 \\ c_1 * SOC - d_1 \quad if \quad SOC \ \geq a_1 \end{cases} \tag{1}$$

$P_s(SOC)$ is the power provided by the model [kW], P_s^* refers to the power set point, i.e., the reference value for the charge (and discharge) [kW], SOC is the state of charge [%], a_1 is the threshold value of the SOC for the transition between the two segments of the piece-wise pattern, constant power and linear decrease power while c_1 and d_1 are parameters characterizing the second segment. These parameters have been estimated by employing the least squares criterion corresponding to the following minimization (2)

$$J = \min_{a,c,d} \sum_{n=1}^{N} [P_{s,n}(SOC_n) - P_s(SOC_n)]^2 \tag{2}$$

$P_{s,n}(SOC_n)$ is the nth measurement of the power flow corresponding to nth measurement of the SOC, whereas $P_s(SOC_n)$ is the power flow computed by the model and of a measured value of SOC_n. The results provided the following values for the parameters: a_1 = 52, c_1 = 0.5765, d_1 = 59.896.

Model for the discharge phase
The model for the discharge phase is more straightforward than those for the charging one. Studying the experimental data, the output from the storage follows the reference for any SOC value, so the model is merely a constant equal to the desired reference power.

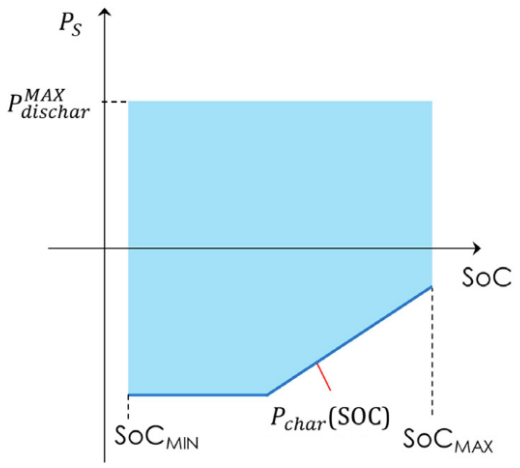


Figure 2: Battery model boundary

3 The architecture of the system

The control architecture is designed to operate in an environment with two exogenous signals: the former, P_D, represents local energy consumption, the latter, P_{RES}, represents the local energy production generated by PV.
To account for real-world uncertainty, it is possible to incorporate unpredictable noise in the exogenous signals:

- Power production noise: which considers different weather conditions, this noise allows for a maximum deviation of +10% above the forecast and -40% below.

- Power consumption noise: which reflects real-world consumption fluctuations, incorporating noise into the power consumption data. This noise introduces a maximum error of ±2.5 kW.

In light of the working environment a bi-level control system has been designed as follows: in the higher level, an optimization problem is developed and solved under an MPC scheme, based on the forecasted exogenous signals. The lower level adjusts the control signals, received from the higher level, in real-time based on the actual values of the exogenous signals. This level is designed with an if-else logic in order to be executed on a PLC. The levels are both implemented and executed in MATLAB as well as the exogenous signals.

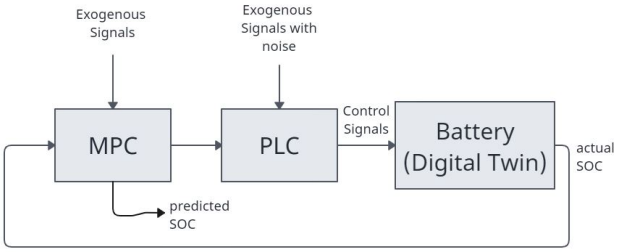


Figure 3: Architecture system scheme

3.1 The higher level optimization problem

The higher level is formulated as a Mixed Linear Integer Programming optimization solved following a receding horizon approach. In particular, only the optimal solution related to the first time interval is used as a control input and then a new run of the optimization model is performed on the next interval updating both the estimated parameters and the real ones. Under a receding horizon approach, the optimization problem is run at the beginning of each fixed interval of time T and only the solution of first step is considered. The objective function is made up of two terms:

$$J = min \sum_{t=0}^{T-1} (\hat{C}_{grid,t}) \tag{3}$$

$\hat{C}_{grid,t}$ is the grid cost [€] and T is the length of the optimization horizon.

$\hat{C}_{grid,t}$ is defined as:

$$\hat{C}_{grid,t} = \begin{cases} C_t P_{grid,t} \Delta & if \quad P_{grid,t} \geq 0 \\ B_t P_{grid,t} \Delta & otherwise \end{cases} \tag{4}$$

with t=0,...,T-1

where C_t is the unit cost to purchase electrical energy from the main grid [€/kWh], $P_{grid,t}$ is the power exchanged with the main grid [kW],Δ is the time discretization interval [h], B_t is the selling price of electrical energy [€/kWh]. Thus the term of the objective function for the grid is given by:

$$\hat{C}_{grid,t} = C_t P_{grid,IN,t} - B_t P_{grid,OUT,t} \tag{5}$$

with t=0,...,T-1

where

$$P_{grid,t} = P_{grid,IN,t} - P_{grid,OUT,t} \tag{6}$$

with t=0,..,T-1

In which, $P_{grid,IN,t}$ and $P_{grid,OUT,t}$ are positive variables indicating power drawn from or sent to the grid.

The SOC of the storage system is given by:

$$x_{t+1} = \alpha x_t - \frac{\eta_t P_{s,t} \Delta}{CAP} \tag{7}$$

$$\eta_t = \begin{cases} \eta_c \;\; if \;\; P_{s,t} \leq 0 \\ 1/\eta_d \;\; otherwise \end{cases} \tag{8}$$

where x_t is SOC of the battery, α is a loss coefficient, η_c is the efficiency in charging mode and η_d is the efficiency in discharging mode. The storage system equation expressed in (7) and (8) is denoted as a mixed logical dynamical (MLD) system [3]. These are described by linear dynamic equations subject to linear inequalities involving real and integer variables [4]. That is:

$$x_{t+1} = \alpha x_t + \frac{(\eta_c - 1/\eta_d) z_{s,t} \Delta}{CAP} - \frac{1}{\eta_d} \frac{P_{s,t} \Delta}{CAP} \tag{9}$$

$$-m_t \lambda_t \leq -P_{s,t} - m_t \tag{10}$$

$$-(M_t + \epsilon_t) \lambda_t \leq P_{s,t} - \epsilon_t \tag{11}$$

$$z_{s,t} \leq M_t \lambda_t \tag{12}$$

$$z_{s,t} > m_t \lambda_t \tag{13}$$

$$z_{s,t} \leq -P_{s,t} - m_t (1 - \lambda_t) \tag{14}$$

$$z_{s,t} \geq -P_{s,t} - M_t (1 - \lambda_t) \tag{15}$$

with t=0,..,T-1

where M_t and m_t are parameters: the first greater than the storage output power limit and the second lower than the storage output power limit [kW], λ_t is a binary auxiliary variable, ϵ_t is a small tolerance (typically the machine precision), $z_{s,t}$ is an auxiliary variable. Of particular importance is the behavior of the battery, so we introduce an additional constraint given by the battery model described in the previous section. In fact, the model allows defining an area in which the values of the power and the SOC are acceptable Fig.2. The identified model allows defining properly the boundary of the area representing the feasible region, which is described by the following constraints:

$$P_{s,char}^{MIN}(SOC) \leq P_{s,t} \leq P_{dischar}^{MAX} \tag{16}$$

$$x_s^{MIN} \leq x_t \leq x_s^{MAX} \tag{17}$$

$$x_{t=0} = SOC_0 \tag{18}$$

In the previous expressions, x_s^{MIN} and x_s^{MAX} (corresponding in the range from 0 to 1 to the SOC_{MIN} and SOC_{MAX} in the range) are the minimum and maximum working limits for the state of charge, $P_{dischar}^{MAX}$ is the maximum power the storage system can deliver when discharging (assumed constant) and P_{char}^{MIN} is the lower limit of the power the storage can absorb when recharging, expressed as a function of SOC, based on the previously identified models, $x_{t=0}$ is the SOC value at the time of execution of the MPC, which represents the actual SOC value, SOC_0, of the battery. In this simulation, not having a real accumulator, the battery behaviour is simulated by a digital twin which receives the input signals computed by the control architecture

The values of the previous parameters are represented in Table 1

Parameter	Value	Parameter	Value
C_t	0.3[€/kWh]	B_t	0.2[€/kWh]
CAP	141 [kWh]	Δ	0.25 [h]
η_c	0.89	α	0.999
η_d	0.89	M_t	50 [kW]
x_S^{MIN}	0.2	m_t	-50 [kW]
x_S^{MAX}	0.9	ϵ_t	0.01

Table 1: System parameters

For the principle of energy conservation, we have:

$$P_{RES,t} + P_{s,t} - P_{D,t} + P_{grid,t} = 0 \tag{19}$$

3.2 The lower level

In this level, the control system's primary function is to compensate for the effects of unpredictable disturbances added to the exogenous signals. The implemented algorithm, coded in MATLAB, is designed to run on a PLC architecture and can be described as follows:

- Whether the state of charge is within the working range $]\overline{SOC}_{MIN};\ \overline{SOC}_{MAX}[$ the goal is to compensate the disturbances using exclusively the accumulator:

$$\begin{cases} P_{s,PLC} = P_{s,t} + [\epsilon_{P_D} - \epsilon_{P_{RES}}] \\ P_{grid,PLC} = P_{grid,t} \end{cases} \tag{20}$$

- Whether the state of charge is higher than $\overline{SOC}_{MAX}$, the goal is to sell the power surplus (ϵ_{P_D}-$\epsilon_{P_{RES}} < 0$) (21) or to discharge the battery to compensate power deficit (ϵ_{P_D}-$\epsilon_{P_{RES}} > 0$) (22):

$$\begin{cases} P_{s,PLC} = P_{s,t} \\ P_{grid,PLC} = P_{grid,t} + [\epsilon_{P_D} - \epsilon_{P_{RES}}] \end{cases} \tag{21}$$

$$\begin{cases} P_{s,PLC} = P_{s,t} + [\epsilon_{P_D} - \epsilon_{P_{RES}}] \\ P_{grid,PLC} = P_{grid,t} \end{cases} \tag{22}$$

- Whether the state of charge is lower than $\overline{SOC}_{MIN}$ the goal is to store the power surplus (23) in the battery or to buy the necessary power to compensate for any power deficit (24):

$$\begin{cases} P_{s,PLC} = P_{s,t} + [\epsilon_{P_D} - \epsilon_{P_{RES}}] \\ P_{grid,PLC} = P_{grid,t} \end{cases} \tag{23}$$

$$\begin{cases} P_{s,PLC} = P_{s,t} \\ P_{grid,PLC} = P_{grid,t} + [\epsilon_{P_D} - \epsilon_{P_{RES}}] \end{cases} \tag{24}$$

Where $P_{s,PLC}$ is the effective power exchange applied to the battery, $P_{s,t}$ is the power exchange computed by the MPC overlooking the disturbances, ϵ_{P_D} is the error between actual and forecasted power consumption, $\epsilon_{P_{RES}}$ is the error between actual and forecasted power production, $P_{grid,PLC}$ is the effective power exchange applied to the grid, $P_{grid,t}$ is the power exchange computed by the MPC overlooking the disturbances. To guarantee feasibility at the next MPC execution, we define a working range for the PLC, which is a proper subset of the SOC feasibility set:

$$]\overline{SOC}_{MIN}; \overline{SOC}_{MAX}[\subset [SOC_{MIN}; SOC_{MAX}] \tag{25}$$

4 Simulations and Results

In order to test the control system, the following exogenous signals were considered Fig.4 and Fig5:

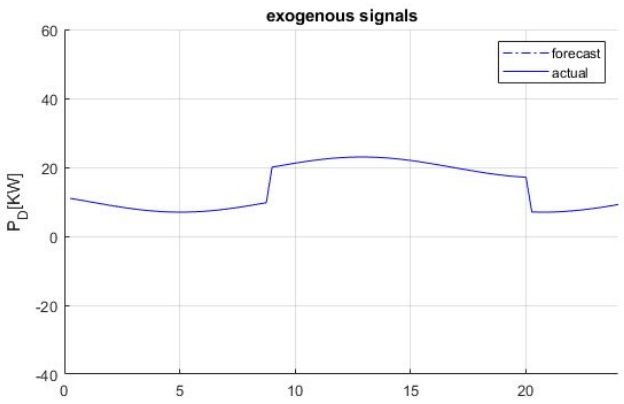


Figure 4: Exogenous signals without noise

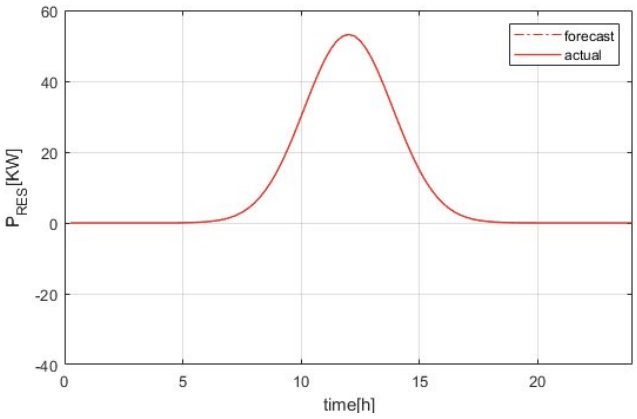


Figure 5: Exogenous signals without noise

The simulations results reveal a control system heavily influenced by the prediction horizon T. This is evident in the SOC graphs, which show scarce utilization of the accumulation system.

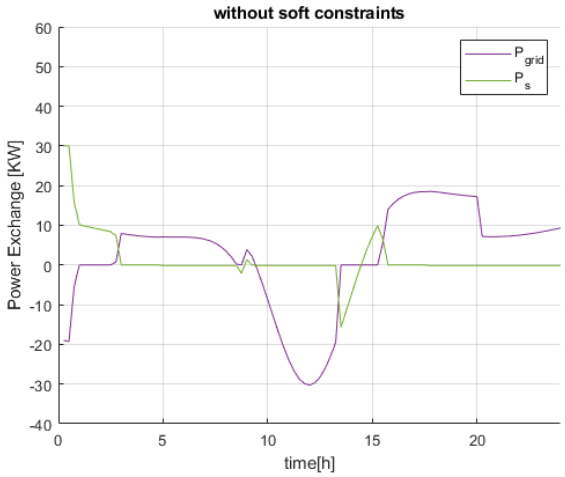


Figure 6: Control signals with T=10

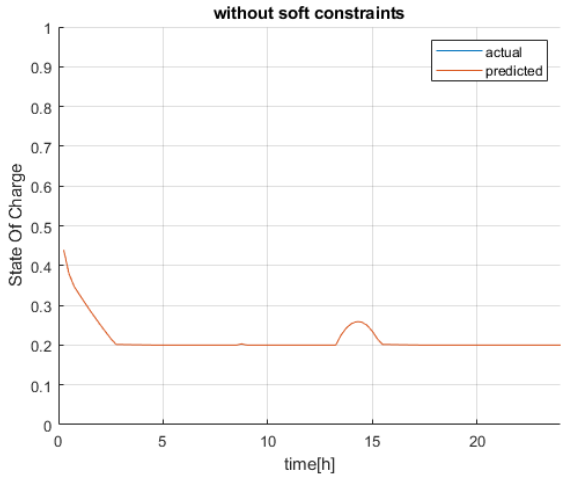


Figure 7: SOC with T=10

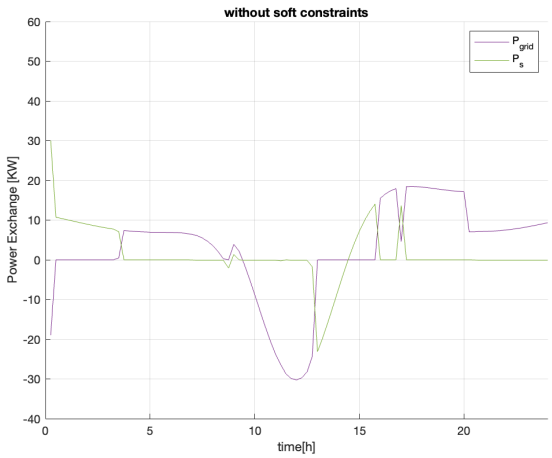


Figure 8: Control signals with T=15

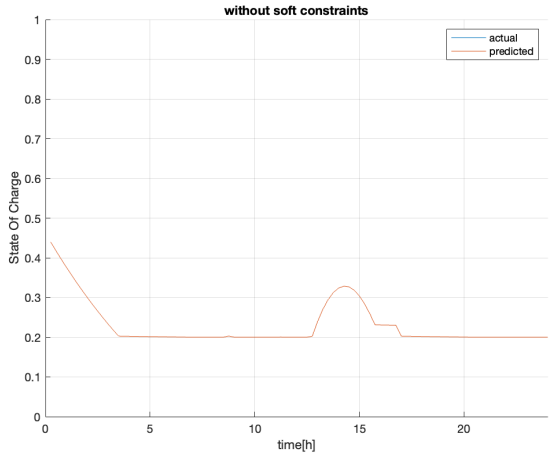


Figure 9: SOC with T=15

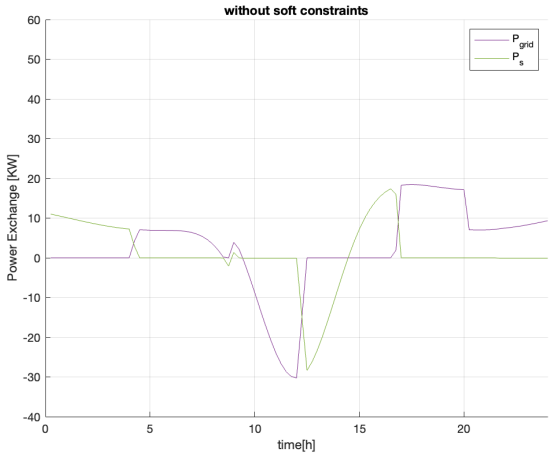


Figure 10: Control signals with T=20

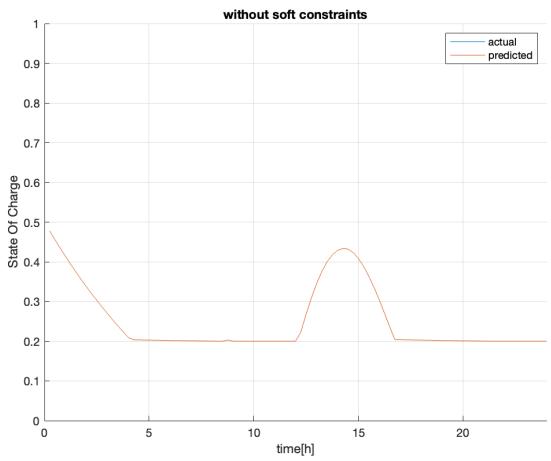


Figure 11: SOC with T=20

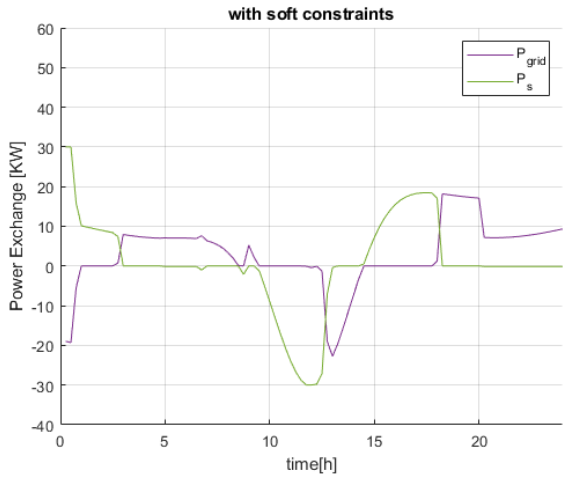


Figure 12: Control signals with soft constraints, T=10

Prediction Horizon	Cost
10	104.26
15	100.20
20	96.00

Table 2: Simulation results without constraints

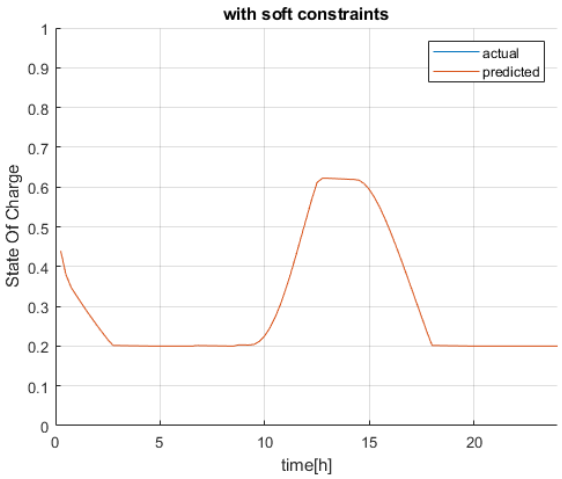


Figure 13: SOC with soft constraints, T=10

The graphs show an increase in the usage of the accumulator as the MPC controller looks farther in the future, thus obtaining a lower global cost, but at the expense of a growing computational burden.

In light of these limitations, the main goal becomes lowering the global cost without excessively increasing the computation time. The solution adopted in this paper consists of introducing a new term in the objective function which rewards the usage of the accumulation system based on exogenous signals:

$$J = min \sum_{t=0}^{T-1} [\hat{C}_{grid,t} - x_t(\zeta_1 P_{RES,t} - \zeta_2 P_{D,t})] \quad (26)$$

Adding this term incentivizes battery charging when forecasted power production increases. Conversely, it rewards accumulator discharge when expected power consumption rises. The results of the simulation are shown in the following graphs:

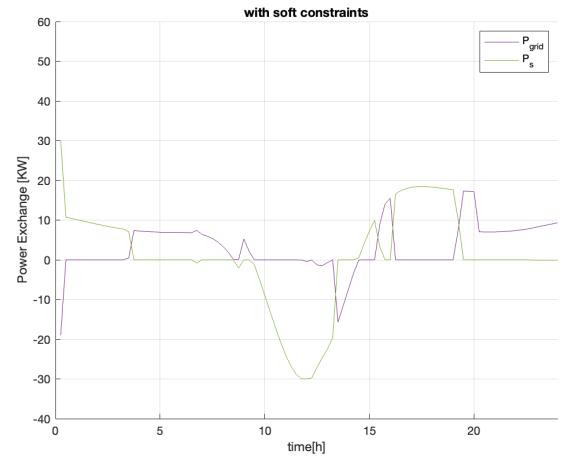


Figure 14: Control signals with soft constraints, T=15

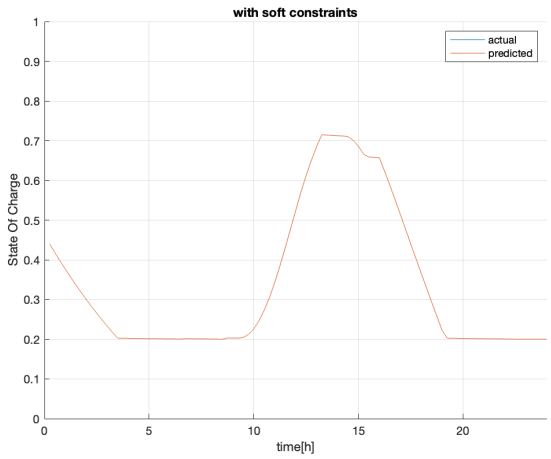


Figure 15: SOC with soft constraints, T=15

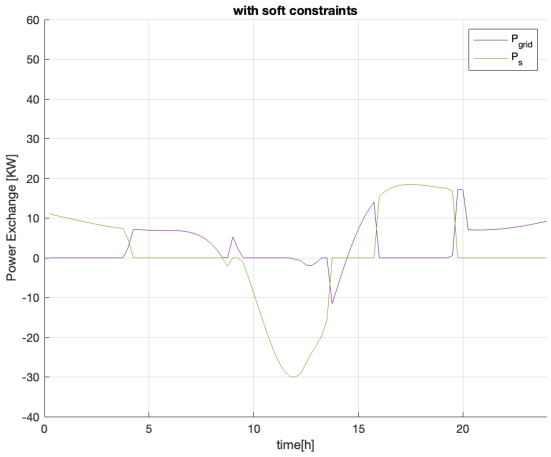


Figure 16: Control signals with soft constraints, T=20

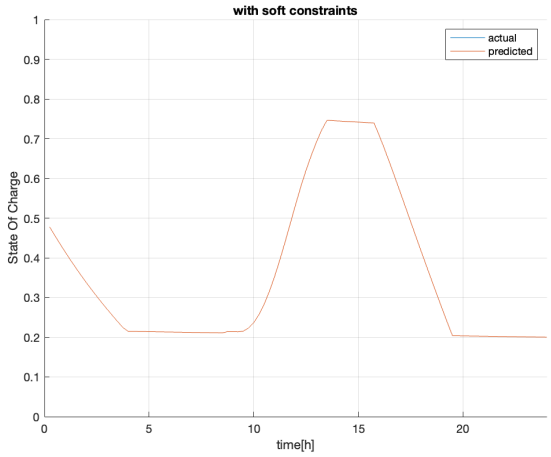


Figure 17: SOC with soft constraints, T=20

Prediction Horizon	ζ_1	ζ_2	Cost
10	0.35	0.03	96.81
15	0.18	0	92.50
20	0.15	0	90.34

Table 3: Simulation results with soft constraints

These results are encouraging, showing that a prediction horizon of T=10 with the soft constraints achieves a global cost nearly equivalent to that obtained with T=20, without using soft constraints tab:1.

In the end it is necessary to point out that the parameters ζ_1 and ζ_2 tuning is based on the exogenous signals' behaviour, in this case ζ_1= 0.35 ζ_2= 0.03 lead to the best results.

As mentioned in section 3, the exogenous signals affected by unforeseeable disturbances are now considered:

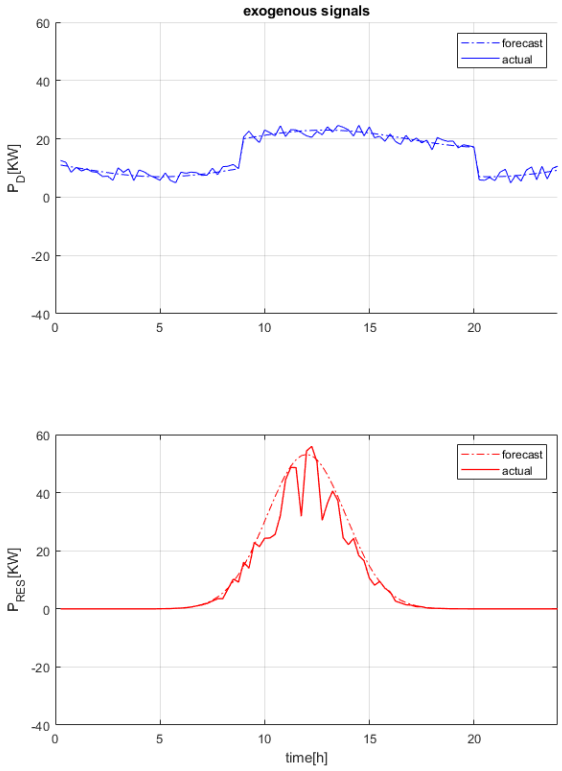


Figure 18: Exogenous signals with noise

The simulation results, represented in Figures [19-22] and represented in the first row of Table 3, compares the performance of the model with and without soft constraints, when subjected to the same noises. As can be seen in Table 3, the model with soft constraints achieves a lower cost compared to the unconstrained model, despite the presence of noise.

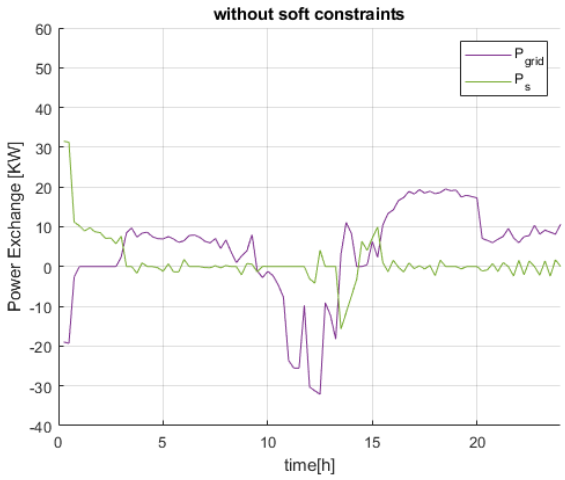


Figure 19: Control signals with noise and without constraints

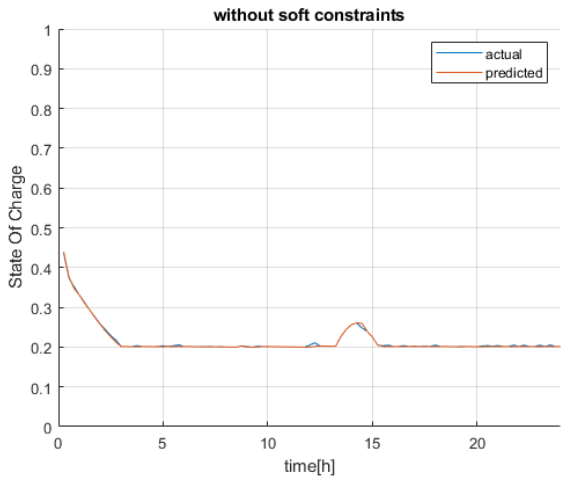


Figure 20: SOC with noise and without constraints

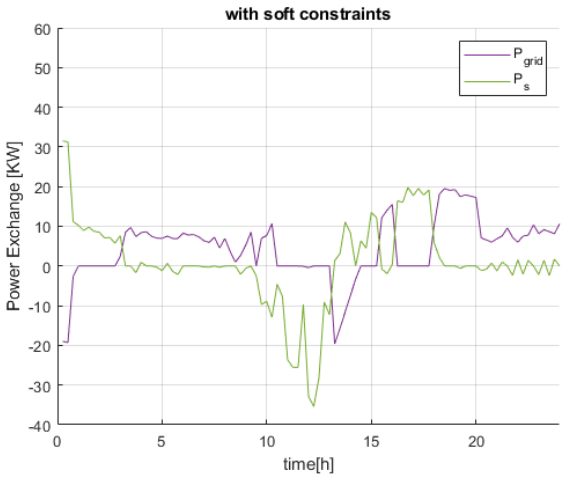


Figure 21: Control signals with noise and with constraints

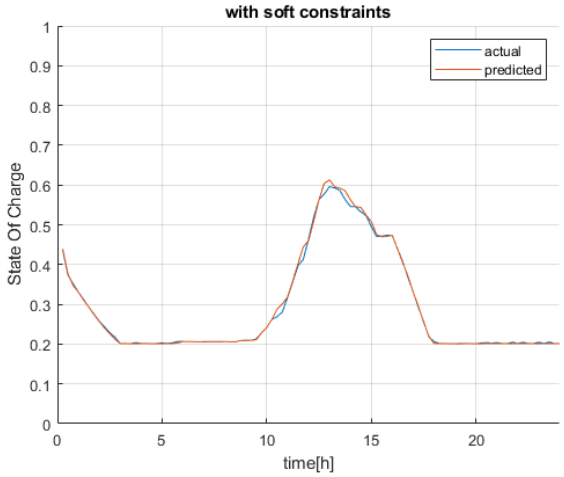


Figure 22: SOC with noise and with constraints

N°Sim	Cost with soft constraints	Cost without soft constraints
1	136.85	141.82
2	124.06	127.22
3	137.13	141.52
4	132.43	136.52

Table 4: Simulation results with noise

The proposed approach is adaptable to various exogenous signal waveforms, potentially leading to performance boost across diverse scenarios after an opportune tuning of ζ_1 and ζ_2 parameters.

Building on the previous analysis, the following graphs Figures [24-27] present the results of further simulations using the input signals in Figure 23. Without using soft constraints, the global cost is 191.00, while adopting ζ_1=0.30 and ζ_2=0.02, it becomes 183.44.

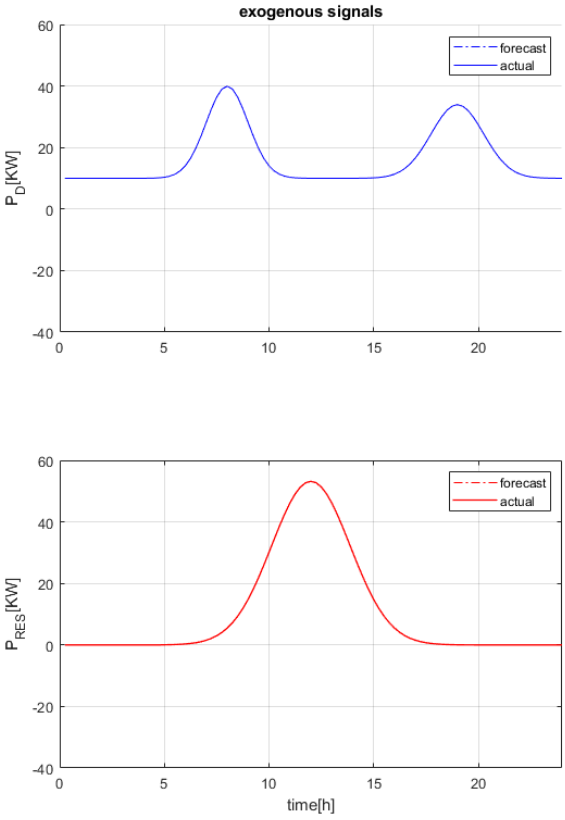


Figure 23: Exogenous signals without noise

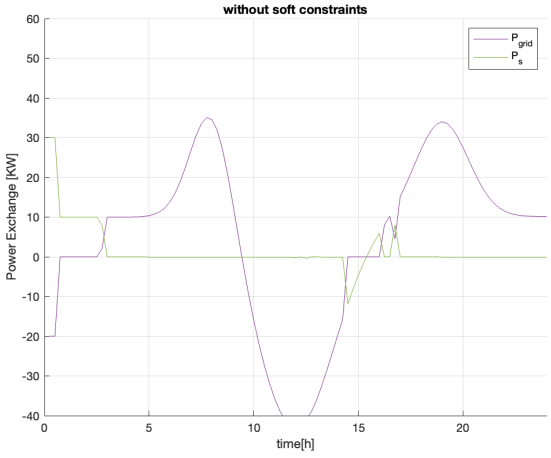


Figure 24: Control signals without noise and without constraints

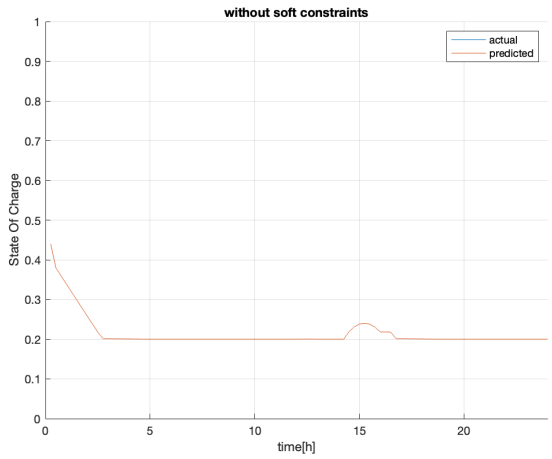


Figure 25: SOC without noise and without constraints

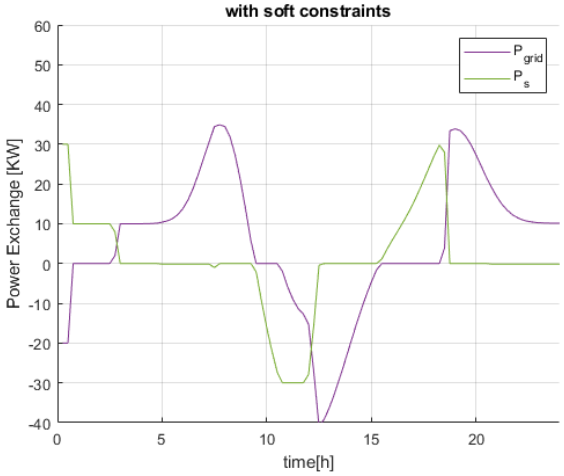


Figure 26: Control signals with constraints but without noise

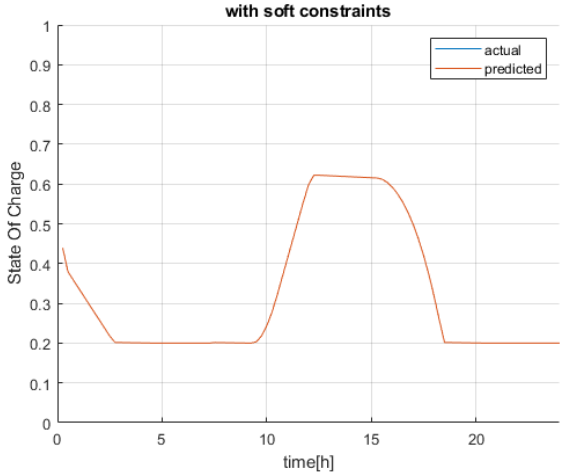


Figure 27: SOC with constraints but without noise

In line with the previous approach, we now incorporate unpredictable disturbances, one for each simulation pair, added directly to the exogenous signals:

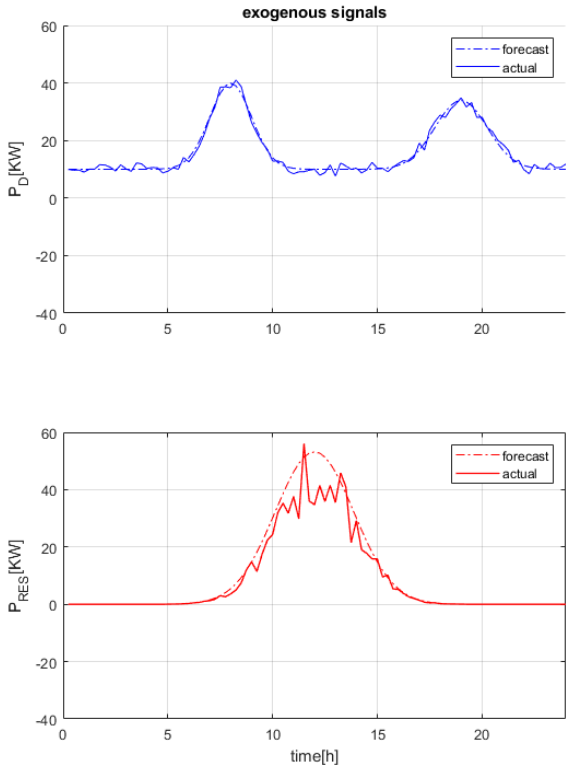


Figure 28: Exogenous signals with noise

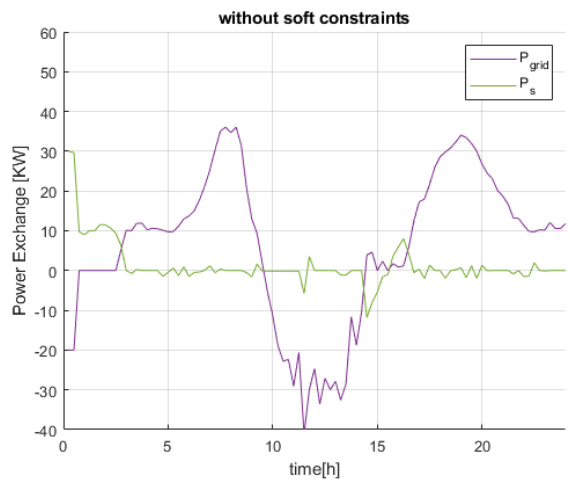


Figure 29: Control signals with noise but without constraints

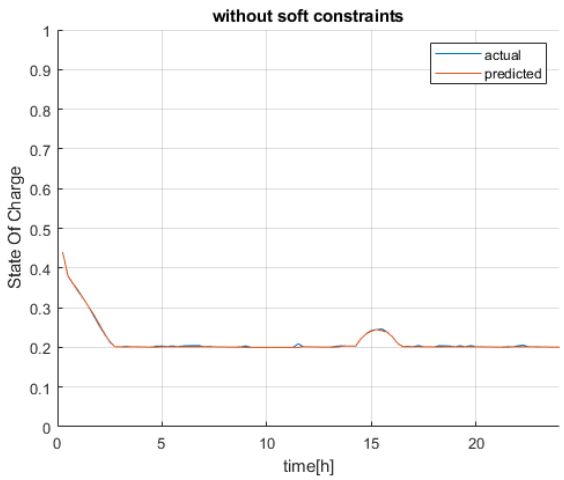


Figure 30: SOC with noise but without constraints

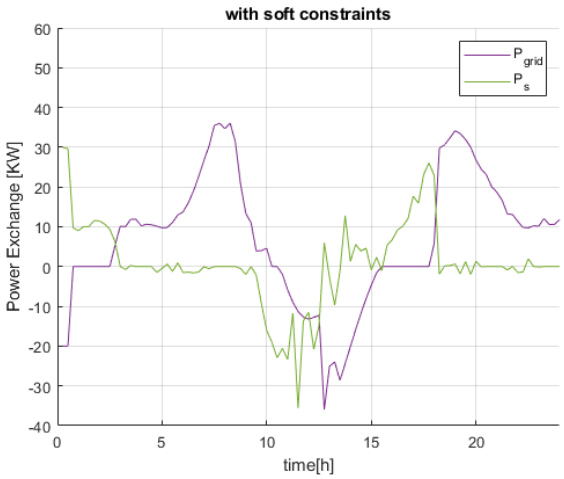


Figure 31: Control signals with constraints and with noise

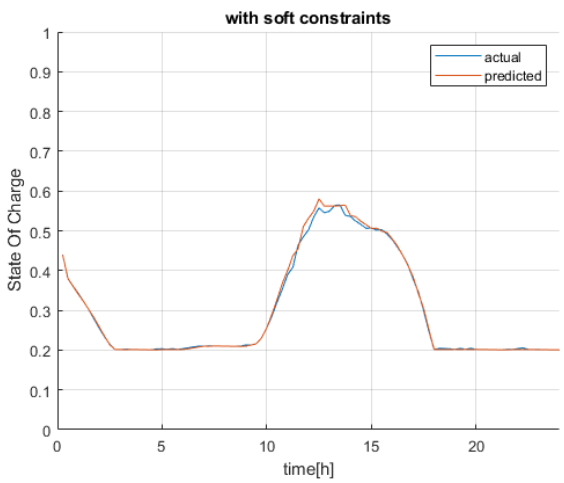


Figure 32: SOC with constraints and with noise

The results of additional simulations are exposed in the Table 5.

N°Sim	Cost with soft constraints	Cost without soft constraints
1	228.55	231.52
2	228.00	229.77
3	220.77	223.50
4	224.86	225.95

Table 5: Simulation results with different load

References

(1) Delfino, F.; Ferro, G.; Minciardi, R.; Robba, M.; Rossi, M.; Rossi, M. Identification and optimal control of an electrical storage system for microgrids with renewables. *Sustainable Energy, Grids and Networks* **2019**, *17*, 100183.

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(4) Parisio, A.; Rikos, E.; Glielmo, L. A model predictive control approach to microgrid operation optimization. *IEEE Transactions on Control Systems Technology* **2014**, *22*, 1813–1827.